



MATHEMATICS METHODS : UNITS 1 & 2, 2020

Test 3 –Trigonometric, Probability, Counting, Exponentials (10%)
(1.2.10 to 1.2.16, 1.3.6 to 1.3.17, 2.1.1 to 2.1.7)

Calculator Assumed - Allow 1 Minute of Reading Time

Time Allowed 25 Minutes	First Name	Surname	Marks 23 marks
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Circle your Teacher's Name:

Bestall

Goh

Fraser-Jones

Freer

Koulianos

Luzuk

Rudland

Tanday

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

6. (1, 3 = 4 marks)

An under 16 soccer coach must choose a team of 11 players from 2 Goalkeepers, 8 Defenders, 6 Midfielders and 6 Strikers.

a. Determine how many different teams of 11 players he can make if there are no restrictions.

b. Determine how many different teams can be made if he has to pick 1 Goalkeeper, 4 Defenders, 3 Midfielders and 3 Strikers to form a team.

7. (2, 2, 2 = 6 marks)

In a small suburb of Perth, it is found that the population of mosquitoes decreases as the population of spiders increases. Scientists found that this can be modelled exponentially.

- a. The population of mosquitoes decreased at a rate of 10% per month and was originally 100,000. Determine an exponential equation in the form $P_m = k(a)^t$ to represent this information.
- b. Determine, after which month the mosquito population will be halved.
- c. If the population of spiders increases by 20% per month and can be modelled according to the exponential equation $P_s = 10000(1.2)^t$, determine when the populations will be equal leaving your answer to one decimal place.

8. (2, 2, 3 = 7 marks)

Given that for events A and B the $P(A) = 0.6$, $P(A|B) = 0.6$ and $P(A \cap B) = 0.3$

a. Determine $P(B|A)$

b. Determine $P(\overline{A \cup B} | \bar{B})$

c. Determine whether A and B are independent, mutually exclusive or other.
Justify your answer.

9. (2, 2, 2 = 6 marks)

In Zombieland you have a 90% chance of becoming a Zombie from a worldwide airborne virus called Zee. The disease takes 1 week for an individual to experience symptoms and start turning into a Zombie. A test to identify the disease has been rushed into production, but it only shows an 80% success rate in detecting those who have the disease and a 90% success rate for those who do not have the disease.

- a. Draw a probability tree to represent this information.
- b. Determine the probability that the test will give the correct result.
- c. Determine the probability that the test is incorrect, despite the test showing they have the disease.

